

Prioritizing Causal Genes at GWAS Loci: Extending PoPS with AlphaGenome Sequence Features

Group 1 • Adrija Chatterjee, Alex Mueller, Yunzhi Han
BDSY 2026 — Genomics Track

July 25, 2026

Abstract

Genome-wide association studies (GWAS) implicate genomic *loci*, but a locus typically contains many genes and rarely names the causal one. The Polygenic Priority Score (PoPS) addresses this by learning, from genome-wide gene features, which genes look like trait-relevant genes; it is a penalised linear model trained on gene-level MAGMA association statistics. We reproduce PoPS as a baseline and ask whether *sequence-based* variant-effect features from AlphaGenome—a DNA sequence-to-function model—add signal beyond PoPS’s gene-level features. We build three models. **Model 1** is a random forest on AlphaGenome features alone; it reaches a pooled top-1-per-credible-set accuracy of 0.200, below MAGMA (0.373) and far below PoPS (0.499). Diagnosing this, we find the limitation is the *model class and feature-integration strategy*, not the sequence features: **Model 2a** replaces the PoPS forest with a ridge regression on the identical PoPS features, nearly doubling accuracy to 0.436 (recovering ~87% of PoPS from a model lacking only PoPS’s gene-gene LD term), and stacking that ridge model with the AlphaGenome forest reaches 0.360. **Model 2b**, a single ridge on the concatenated feature matrix, reaches 0.403, statistically indistinguishable from ridge-on-PoPS-alone (0.401): once a strong linear model has the PoPS features, AlphaGenome is largely redundant *in the pooled metric* (adding it helps 6 traits, hurts 6, ties 33). Yet the sequence signal is real and trait-specific: on a coherent class of quantitative serum biomarkers (total protein, total cholesterol, urate, albumin, SHBG) our models *exceed* PoPS, and the learned stacking weights independently load on AlphaGenome exactly for lipid, blood-cell and endocrine traits. Requiring our models to agree with PoPS or MAGMA (an intersection rule) raises precision substantially (to 78% when ridge and MAGMA agree) at the cost of recall. We conclude that model class dominated performance, that AlphaGenome’s value is real but concentrated on traits with sparse, strong, local cis-regulatory architecture, and that per-trait-adaptive fusion is the clearest next step.

Contents

1	Introduction: concepts and tools	2
1.1	From variants to loci to genes	2
1.2	MAGMA: gene-level association	2
1.3	PoPS: the Polygenic Priority Score	3
1.4	AlphaGenome: sequence to function	3
1.5	CADD: a deleteriousness prior	3
1.6	Fine-mapping, credible sets, and the gold standard	3
1.7	Random forests and ridge regression	3
2	Problem formulation	3

3	Data	4
4	Methods	4
4.1	AlphaGenome feature construction	4
4.2	Target construction	4
4.3	Model 1 — random forest on AlphaGenome	5
4.4	Model 2a — ridge on PoPS, forest on AlphaGenome, stacked (late fusion)	5
4.5	Model 2b — ridge on concatenated features (early fusion)	5
4.6	Penalty selection	5
4.7	Intersection with PoPS and MAGMA	5
4.8	Evaluation protocol	5
5	Limitations and how we addressed them	6
6	Hyperparameters	6
7	Results	6
7.1	Model class, not features, was the bottleneck	6
7.2	AlphaGenome is redundant in the pooled metric under a linear model	7
7.3	Integration strategy matters	7
7.4	Which traits load on AlphaGenome	8
7.5	Beating PoPS on a coherent biomarker class	8
7.6	Intersection with PoPS and MAGMA yields a high-precision subset	8
7.7	Feature importance: AlphaGenome is over-represented among top features for biomarker traits	8
8	Discussion	11
9	Future scope	11

1 Introduction: concepts and tools

1.1 From variants to loci to genes

A single-nucleotide polymorphism (SNP) is a common single-base difference between genomes. GWAS test statistical association between SNPs and a trait. Because nearby variants are correlated through *linkage disequilibrium* (LD), a significant GWAS signal identifies a genomic *locus*—a region—rather than the exact causal variant or gene. Two complementary lines of evidence help resolve the causal gene: *gene-level association* (which genes carry the aggregate signal) and *functional* evidence (what nearby variants do to which gene).

1.2 MAGMA: gene-level association

MAGMA [1] aggregates SNP-level GWAS statistics within a gene, correcting for LD, and returns a per-gene *z*-statistic **ZSTAT** quantifying how strongly each gene associates with the trait genome-wide. Throughout, **ZSTAT** is the regression *target* every model is trained to predict. Crucially, **ZSTAT** is an *association* signal (high for many genes near a strong locus), not a clean causal label.

1.3 PoPS: the Polygenic Priority Score

PoPS [2] learns which genome-wide gene features (bulk and single-cell expression, biological pathways, protein–protein interaction networks) predict the MAGMA signal, then scores each gene at a locus by how well it matches the learned, trait-specific profile. PoPS is a *penalised linear model* fit on gene features with a gene–gene covariance (LD) term; it is the framework we reproduce and extend.

1.4 AlphaGenome: sequence to function

AlphaGenome [3] maps a DNA sequence (up to ~ 1 Mb) through sequence encoders, transformers, and task-specific decoders to thousands of functional readouts across cell types—expression, chromatin state, transcription-factor binding, splicing, contacts. We use its predicted effect of a variant on RNA abundance (Δ RNA) across 285 ontology contexts. Whereas PoPS describes *what a gene is*, AlphaGenome describes *what nearby variants do*.

1.5 CADD: a deleteriousness prior

CADD [4] scores each variant’s predicted deleteriousness; its PHRED-scaled form is a genome-wide rank score (e.g. 20 = top 1%, 30 = top 0.1% most deleterious). We use CADD to up-weight likely-functional variants when aggregating AlphaGenome effects to genes.

1.6 Fine-mapping, credible sets, and the gold standard

Statistical fine-mapping refines a locus into one or more *credible sets*: small groups of variants that collectively contain the causal variant with high posterior probability (e.g. 95%). A single locus can harbour *several* independent causal signals, each its own credible set (hence a credible-set ID distinct from the locus ID). Our validation set [8] restricts to the 1,348 credible sets that each contain a fine-mapped *coding* variant. A coding variant sits inside one gene’s protein-coding sequence, so it points *unambiguously* to its gene; that gene is *taken as* the causal gene g^* for the credible set—a high-confidence, orthogonal label the models never see. Credible sets with no coding variant have no trustworthy label and are excluded.

1.7 Random forests and ridge regression

A random forest [5] averages many regression trees, each fit on a bootstrap resample using a random feature subset per split; it is non-parametric, captures non-linearity, and is robust to correlated features. Ridge regression [6] minimises squared error plus an ℓ_2 penalty $\lambda \|\beta\|_2^2$, shrinking coefficients to control variance and multicollinearity when the number of features p exceeds the number of samples n —the regime of the PoPS block.

2 Problem formulation

Let genes be indexed $j \in \mathcal{G}$. For each gene we observe a PoPS feature vector $\mathbf{x}_j^{\text{PoPS}} \in \mathbb{R}^{57,742}$, an AlphaGenome feature vector $\mathbf{x}_j^{\text{AG}} \in \mathbb{R}^{285}$, and a regression target $y_j = \text{ZSTAT}_j \in \mathbb{R}$. We learn a scoring map $s : \mathbb{R}^d \rightarrow \mathbb{R}$ such that, within each credible set L (with candidate genes $\mathcal{G}_L \subseteq \mathcal{G}$), the top-scoring gene is the predicted causal gene:

$$\hat{g}_L = \arg \max_{j \in \mathcal{G}_L} s(\mathbf{f}_j), \quad \text{Acc} = \frac{1}{|\mathcal{L}|} \sum_{L \in \mathcal{L}} \mathbb{1}[\hat{g}_L = g_L^*], \quad (1)$$

where $\mathbf{f}_j$ is the feature block fed to s (PoPS, AlphaGenome, or their concatenation; dimension $d \leq 58,027$), $\mathcal{L}$ is the set of 1,348 credible sets, and g_L^* is the coding-variant-labelled causal gene. Because each credible set is scored with a single prediction and has a single positive gene, top-1 accuracy equals precision equals recall equals F_1 under this protocol.

3 Data

Table 1: Datasets and dimensions. All are from the Finucane Lab data release [8] except the AlphaGenome features, which we computed [3].

Dataset	Dimensions	Source
PoPS gene features	15,795 genes $\times$ 57,742 features	Finucane Lab
AlphaGenome Δ RNA features	9,669 genes $\times$ 285 ontologies	computed (AlphaGenome)
MAGMA gene scores (ZSTAT, target)	112 traits, $\sim$ 15,795 genes	Finucane Lab
Coding credible-set gold standard	1,348 credible sets (589 positive genes)	Finucane Lab

AlphaGenome features were computed from 192,266 per-variant predictions (7.66×10^9 variant–gene–ontology rows) and aggregated to $9,669 \times 285$. Only $\sim 21\%$ of AlphaGenome variants had a matching CADD score. The combined feature matrix has $57,742 + 285 = 58,027$ columns. Genes present in PoPS but not AlphaGenome (15,795 vs. 9,669) receive a mean-imputed, neutral AlphaGenome block.

4 Methods

4.1 AlphaGenome feature construction

For gene j , let $\{i\}$ index SNPs within 1 Mb of its TSS. Each SNP i has an AlphaGenome Δ RNA vector $\mathbf{v}_i \in \mathbb{R}^{285}$, a distance d_i to the gene *body* [start, end], and a CADD PHRED score C_i . We weight each SNP by proximity and deleteriousness and aggregate by a weighted *sum*:

$$w_{ij} = \underbrace{\begin{cases} 1, & d_i = 0 \text{ (inside body)} \\ \exp(-d_i/\tau), & d_i > 0 \end{cases}}_{\text{distance}} \times \underbrace{(C_i/c)}_{\text{CADD}}, \quad \mathbf{x}_j^{\text{AG}} = \sum_i w_{ij} \mathbf{v}_i \in \mathbb{R}^{285}, \quad (2)$$

with decay length $\tau = 100$ kb and CADD scale $c = 1.0$. A single variant contributes to *every* gene whose 1 Mb window contains it, with a per-gene weight, so it loads most strongly on the gene(s) it lies in or nearest. Variants lacking CADD are dropped; a gene with no usable variant receives a neutral (mean-imputed) feature.

4.2 Target construction

The target y_j is the MAGMA ZSTAT residualised on confounders, PoPS-style:

$$y_j = \text{ZSTAT}_j - \mathbf{c}_j^\top \hat{\boldsymbol{\beta}}_c, \quad \mathbf{c}_j = (\log \ell_j, \log m_j, \rho_j), \quad (3)$$

where ℓ_j is gene length, m_j the SNP count, and $\rho_j = m_j/\ell_j$ the SNP density; $\hat{\boldsymbol{\beta}}_c$ is the OLS fit of ZSTAT on $[1, \mathbf{c}_j]$.

4.3 Model 1 — random forest on AlphaGenome

A random forest [5] regresses y_j on a feature block. We evaluate it on AlphaGenome ($\mathbf{X}^{\text{AG}}$), PoPS ($\mathbf{X}^{\text{PoPS}}$), and their concatenation ($\mathbf{X}^{\text{comb}}$). Held-out scores are out-of-bag (OOB): a gene is scored only by the trees that did not bootstrap it.

4.4 Model 2a — ridge on PoPS, forest on AlphaGenome, stacked (late fusion)

We learn one view from PoPS with the estimator PoPS itself uses—ridge regression,

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}^{\text{PoPS}} \beta\|_2^2 + \lambda \|\beta\|_2^2 = (\mathbf{X}^{\text{PoPS}\top} \mathbf{X}^{\text{PoPS}} + \lambda \mathbf{I})^{-1} \mathbf{X}^{\text{PoPS}\top} \mathbf{y}, \quad \hat{r}_j = \mathbf{X}_j^{\text{PoPS}} \hat{\beta}(\lambda), \quad (4)$$

producing held-out PoPS scores $\hat{r}_j$ by 5-fold out-of-fold prediction, and one view from AlphaGenome via the Model-1 random forest ($\hat{a}_j$, OOB). We standardise each block score ($z(\cdot)$; genes missing AlphaGenome take the block mean) and fuse them with a *ridge* meta-learner (late fusion / stacking [7]):

$$\hat{y}_j = \gamma_0 + \gamma_1 z(\hat{r}_j) + \gamma_2 z(\hat{a}_j), \quad \gamma = \arg \min_{\gamma} \sum_{j \in \text{train}} (y_j - \hat{y}_j)^2 + \eta \|(\gamma_1, \gamma_2)\|_2^2. \quad (5)$$

The meta-learner is ridge rather than ordinary least squares because the two block scores are correlated (both predict y_j); the ℓ_2 penalty η , tuned by cross-validation, shrinks the fusion weights rather than fitting them unconstrained.

4.5 Model 2b — ridge on concatenated features (early fusion)

Alternatively we fuse at the feature level: concatenate the blocks and fit one ridge,

$$\mathbf{X}^{\text{comb}} = [\mathbf{X}^{\text{PoPS}} \mid \mathbf{X}^{\text{AG}}] \in \mathbb{R}^{n \times 58,027}, \quad \hat{s}_j = \mathbf{X}_j^{\text{comb}} \hat{\beta}, \quad (6)$$

with the AlphaGenome block *standardised before concatenation* so a single global penalty λ is scale-fair (Section 5).

4.6 Penalty selection

All ridge penalties are chosen by generalised cross-validation (GCV, an efficient leave-one-out approximation) over $\lambda \in \{10^{-2}, \dots, 10^{10}\}$ (25 log-spaced values), *on the MAGMA regression over the training genes*—never on the credible-set labels.

4.7 Intersection with PoPS and MAGMA

Finally we combine each of our models with the PoPS and MAGMA priors under an intersection rule: a gene is predicted causal for a credible set *iff both* the model and the prior rank it top-1. Requiring agreement raises precision (we commit only when confident) at the cost of recall (we abstain when the two disagree), tracing a point above the precision–recall diagonal.

4.8 Evaluation protocol

We evaluate top-1 per *credible set* (Eq. 1) over the 1,348-set gold standard, comparing trait-by-trait to the original PoPS. Two properties make the comparison fair: (i) the label (g_L^* , from an independent coding variant) is orthogonal to every feature; and (ii) held-out scoring (OOB for forests, out-of-fold for ridge) ensures no gene is scored by a model trained on it. We use full-data

held-out scoring rather than leave-one-chromosome-out (`loco: false`): within a credible set all candidate genes share a chromosome, so any residual chromosome-level information applies equally to all and cannot tilt the top-1 choice.

5 Limitations and how we addressed them

Several difficulties shaped the design; we record each and our response.

Feature-scale imbalance in early fusion. Un-standardised, the AlphaGenome block enters the shared ridge penalty on a different scale from the PoPS features and dominates the fit; our first `ridge_all` run collapsed to 0.163. *Fix:* z -score the AlphaGenome block before concatenation, restoring 0.403.

Block swamping in early-fusion forests. In the 58,027-column matrix, the 57,742 PoPS columns swamp the 285 AlphaGenome columns under a 5% per-split feature fraction, so the combined forest barely exceeds PoPS alone. *Fix:* late-fusion stacking, which fuses one prediction per block on equal footing.

Memory: a $\sim 10^9$ -cell feature matrix. The 20 GB PoPS text cannot be parsed repeatedly. *Fix:* a Feather cache parsed once, reused across all runs.

Correlated base scores in stacking. The two block scores are not independent out-of-sample (shared training genes and target), so an unconstrained OLS meta-learner would over-trust one block. *Fix:* a ridge (regularised) meta-learner; non-negative or per-trait weights are noted as further improvements.

Coding-label imperfection. The gold standard assumes the fine-mapped coding variant is causal and acts on the gene it lies in; a coding variant merely in LD with the true (non-coding) causal variant would mislabel the gene. *Mitigation:* independent signals are separated into distinct credible sets, and any residual label noise is roughly equal across methods, so relative comparisons remain fair.

Partial CADD coverage. Only $\sim 21\%$ of AlphaGenome variants had a CADD score; the rest are dropped from the weighting. This is a coverage limitation we note rather than fully resolve.

6 Hyperparameters

Under `max_features= 0.05`, a split considers ≈ 14 of the 285 AlphaGenome features and $\approx 2,900$ of the 57,742 PoPS features. The forest is deliberately regularised for tractability, so its scores are best read as a conservative lower bound.

7 Results

Our evaluation harness reproduces the published PoPS top-1 accuracy (~ 0.50), validating the protocol. Table 3 gives pooled top-1-per-credible-set accuracy for all models and reference methods.

7.1 Model class, not features, was the bottleneck

Model 1 (random forest) reaches only 0.200–0.254 across feature blocks, below MAGMA. Replacing the PoPS forest with a ridge regression on the *identical* PoPS features (`ridge_pops`) nearly doubles

Table 2: Configuration used throughout (from `config.example.yaml`).

Stage	Parameter	Value
Feature window	variants per gene	± 1 Mb of TSS
Distance weight	decay length τ ; inside-body weight	100 kb; 1.0
CADD weight	transform; scale c	PHRED; 1.0
Aggregation	method	weighted <i>sum</i> (PIP off)
Target	residualise ZSTAT on	log length, log #SNPs, density
Random forest	trees B ; <code>max_features</code> ; <code>min_samples_leaf</code>	100; 0.05; 20
Ridge	λ grid; selection; intercept	$\{10^{-2}, \dots, 10^{10}\}$; GCV; centred
Stacking meta-learner	estimator; weights	ridge; ℓ_2 -shrunk, CV-tuned
Screening	marginal feature pre-selection	none
Held-out scoring	forest; ridge; LOCO	OOB; out-of-fold; false

Table 3: Pooled top-1-per-credible-set accuracy. n is the number of evaluable credible sets for that mode (modes scoring the PoPS gene universe evaluate 1,312; others 1,348). TP/FP are true/false positives. Reference methods italicised.

	Model	n	TP	FP	Top-1 acc.
Model 2a	<code>ridge_pops</code> (ridge on PoPS)	1,312	572	740	0.436
Model 2b	<code>ridge_all</code> (ridge on PoPS+AG)	1,154	—	—	0.403
	<code>ridge_pops_only</code> (same script, PoPS)	1,154	—	—	0.401
Model 2a	<code>stacked</code> (ridge-PoPS + RF-AG)	1,312	472	840	0.360
Model 1	<code>combined</code> (RF on PoPS+AG)	1,348	343	1,005	0.254
Model 1	<code>pops</code> (RF on PoPS)	1,312	318	994	0.242
Model 1	<code>alphagenome</code> (RF on AG)	1,348	269	1,079	0.200
<i>Reference</i>	<i>PoPS (paper)</i>	1,348	—	—	0.499
<i>Reference</i>	<i>MAGMA</i>	1,347	—	—	0.373
<i>Reference</i>	<i>SMR</i>	1,346	—	—	0.287
<i>Reference</i>	<i>Random</i>	1,348	—	—	0.112

accuracy, from 0.242 to **0.436** —recovering $0.436/0.499 \approx 87\%$ of true PoPS from a model that lacks only PoPS’s gene–gene LD term. This is direct evidence that the model class, not the PoPS features, limited Model 1, consistent with the PoPS paper’s own finding that a linear model was best.

7.2 AlphaGenome is redundant in the pooled metric under a linear model

Compared fairly on the same script and loci, `ridge_all` (0.403, PoPS+AG) essentially ties `ridge_pops_only` (0.401, PoPS): adding AlphaGenome to a strong linear model changes pooled accuracy by +0.002. Per trait, AlphaGenome helps 6 traits, hurts 6, and ties 33 (median 0). Its RF-era “advantage” was an artifact of the weak forest baseline. The scale of the effect depended entirely on standardisation: the un-standardised concatenation collapsed to 0.163.

7.3 Integration strategy matters

Late-fusion stacking (0.360) exceeds the early-fusion `combined` forest (0.254) and every other forest variant, but sits below `ridge_pops` (0.436): the meta-learner loads heavily on AlphaGenome (Section 7.4), which helps the trait classes where sequence dominates but dilutes the strong linear-PoPS

signal on the majority of traits. A single globally fixed fusion weight is therefore itself suboptimal.

7.4 Which traits load on AlphaGenome

On the z -score scale of Eq. 5, the AlphaGenome weight exceeds the PoPS weight for most traits, and the traits with the *largest* AlphaGenome weight are almost exclusively molecular blood biomarkers (Table 4): platelet and red-cell indices, lipids, and endocrine/renal measures. That a data-driven meta-learner independently rediscovers this division of labour is evidence the AlphaGenome signal is real rather than an artifact.

Table 4: Traits with the largest learned AlphaGenome stacking weight w_{AG} , with the corresponding PoPS (ridge) weight, on the z -score scale.

Trait	w_{PoPS}	w_{AG}	Trait	w_{PoPS}	w_{AG}
Plt (platelets)	0.45	0.98	SHBG	0.31	0.89
HDLC	0.29	0.95	HbA1c	0.37	0.89
MCH	0.46	0.92	IGF1	0.31	0.89
ApoA	0.29	0.91	RBC	0.42	0.88
MCV	0.47	0.89	ApoB	0.26	0.84

7.5 Beating PoPS on a coherent biomarker class

Although pooled accuracy trails PoPS (0.40 vs. 0.50), our models *exceed* PoPS on a coherent set of traits. Per trait, `ridge_all` beats PoPS on Albumin, Glucose, Total Protein, Urate, ALT, Total Cholesterol, LDLC, SHBG and eBMD; `stacked` beats PoPS on Total Protein, SHBG, Urate, Albumin, Total Cholesterol, IGF1, TG and RBC. The overlap—Total Protein, Total Cholesterol, Urate, Albumin, SHBG—is a clean class of quantitative serum biomarkers with sparse, strong, *local* cis-regulatory architecture, exactly where a sequence-to-expression model carries decisive signal.

7.6 Intersection with PoPS and MAGMA yields a high-precision subset

Requiring our models to agree with an orthogonal prior (Section 4.7) trades recall for precision (Table 5, Fig. 1). Each single method lies on the precision–recall diagonal (precision = recall = top-1 accuracy); every intersection lies well above it. When our ridge model and MAGMA *agree*, the shared gene is causal 78.1% of the time—versus 37.3% for MAGMA alone—though at the cost of committing to a gene on only 19% of credible sets. `Ridge` $\cap$ `PoPS` gives the best balance (precision 0.628 at recall 0.284). This reproduces, quantitatively, the group’s earlier observation that intersecting gene sets raises precision to $\sim 77\%$. The correct reading is that intersection identifies a *high-confidence subset*, not a uniformly better predictor: F_1 falls relative to the baselines because precision is bought with recall.

7.7 Feature importance: AlphaGenome is over-represented among top features for biomarker traits

Because ridge spreads weight across correlated features, we rank importance with a *sparse* model: features are screened by marginal correlation and an elastic net selects a small set (`ridge_coefs.py`). For Albumin—a serum biomarker on which `ridge_all` beats PoPS—the elastic net retains 159 of 58,027 features, of which 13 (8.2%) are AlphaGenome, against a 0.5% baseline share: a $\sim 17\times$ enrichment. The effect concentrates at the very top: the *four most significant features are all*

Table 5: Intersection (“AND”) of each model with an orthogonal prior: a gene is predicted only where both rank it top-1. “Committed” is the number of credible sets on which a prediction is made (of 1,348). Baselines commit on all sets, so precision = recall.

Method	Recall	Precision	F_1	Committed
<i>Original PoPS</i>	0.499	0.499	0.499	1,348
<i>Original MAGMA</i>	0.373	0.373	0.373	1,347
Ridge $\cap$ MAGMA	0.151	0.781	0.252	260
Ridge $\cap$ PoPS	0.284	0.628	0.391	610
Stacked $\cap$ MAGMA	0.108	0.697	0.186	208
Stacked $\cap$ PoPS	0.213	0.629	0.318	456

AlphaGenome, and 10 of the top 50 are *AlphaGenome*—a $\sim 41\times$ over-representation relative to chance (Table 6, Fig. 2); the ten highest-weight features are listed in Table 7.

This refines the earlier “redundant in the pooled metric” result (Section 7.2). Pooled across all 112 traits, adding *AlphaGenome* barely moves top-1 accuracy; but on the specific biomarker traits where it helps, *AlphaGenome* features are not marginal—they carry the *largest* weights in the fitted model. Feature importance is therefore strongly trait-dependent: a highly polygenic trait would be expected to show *AlphaGenome* near its chance share, whereas a local-cis biomarker like Albumin shows it dominating the top of the ranking. The two observations are consistent—a few high-weight sequence features sharpen the ranking on biomarker traits without changing the pooled top-1 accuracy across the full, heterogeneous trait panel.

Table 6: Block composition of the most significant Albumin features (elastic net, $|\text{coef}|$ -ranked). “Expected” is the count under random selection ($N \times 285/58,027$). *AlphaGenome* is 0.5% of features but dominates the top.

Top- N	<i>AlphaGenome</i> (observed)	<i>AlphaGenome</i> (expected)	Enrichment
50	10	0.25	$\sim 41\times$
100	11	0.49	$\sim 22\times$
250	13	1.23	$\sim 11\times$
500	13	2.46	$\sim 5.3\times$
1000	13	4.91	$\sim 2.6\times$

Extending to the top 50 features sharpens the picture. Ten of the fifty are *AlphaGenome* (20%), against 0.25 expected by chance—a $\sim 41\times$ enrichment—and they concentrate toward the very top (4 of the top 10, thinning to 20% by rank 50). Their ontology namespaces are informative: *seven of the ten are UBERON* (tissue/anatomy) contexts, versus two cell-type (CL) and one experimental-factor (EFO) context (Table 8). The sequence signal therefore enters predominantly through *tissue-level* regulatory context rather than fine cell types. Notably the implicated tissues are *diverse*—brain (UBERON:0000955), heart (UBERON:0000948), kidney—and not the canonical albumin-producing liver. This reinforces that the result is a *block-level* statistical signal (*AlphaGenome* features rise to the top as a group) rather than a single-tissue mechanistic claim: coefficient magnitudes on standardised, correlated features identify *which block* carries signal robustly, but the identity of the specific tissue within a correlated group is not itself interpretable.

Table 7: The ten most significant features for Albumin (elastic net on the combined matrix, ranked by |coefficient|). The four highest-weight features are all AlphaGenome regulatory ontologies; PoPS tissue/pathway features follow. UBERON/CL/EFO codes are glossed to their tissue or cell type.

Rank	Feature	Coef.	Block / gloss
1	AG:UBERON:0000955	+0.092	AlphaGenome — brain
2	AG:CL:0002319	+0.084	AlphaGenome — neural cell
3	AG:UBERON:0018118	+0.065	AlphaGenome — neural tube / CNS
4	AG:UBERON:0000948	+0.041	AlphaGenome — heart
5	pops__human_lung_projected_icaloadings.35	−0.039	PoPS — lung expression
6	pops__human_brain2_projected_icaloadings.34	+0.037	PoPS — brain expression
7	pops__human_brain2_projected_icaloadings.19	−0.032	PoPS — brain expression
8	pops__mouse_development_diffexprs_tstat.27	+0.032	PoPS — developmental DE
9	pops__pathways.8375	+0.031	PoPS — pathway membership
10	pops__human_liver_diffexprs_tstat.10	−0.030	PoPS — liver expression

Table 8: The ten AlphaGenome features among the top 50 Albumin features, with ontology namespace. UBERON = anatomy/tissue, CL = cell type, EFO = experimental factor. Tissue (UBERON) contexts dominate (7/10).

Rank	Ontology code	Namespace
1	UBERON:0000955 (brain)	UBERON (tissue)
2	CL:0002319 (neural cell)	CL (cell)
3	UBERON:0018118	UBERON (tissue)
4	UBERON:0000948 (heart)	UBERON (tissue)
14	UBERON:0004538	UBERON (tissue)
17	EFO:0030018	EFO (exp. factor)
20	CL:0000047	CL (cell)
31	UBERON:0001072	UBERON (tissue)
33	UBERON:0008971	UBERON (tissue)
49	UBERON:0015143	UBERON (tissue)

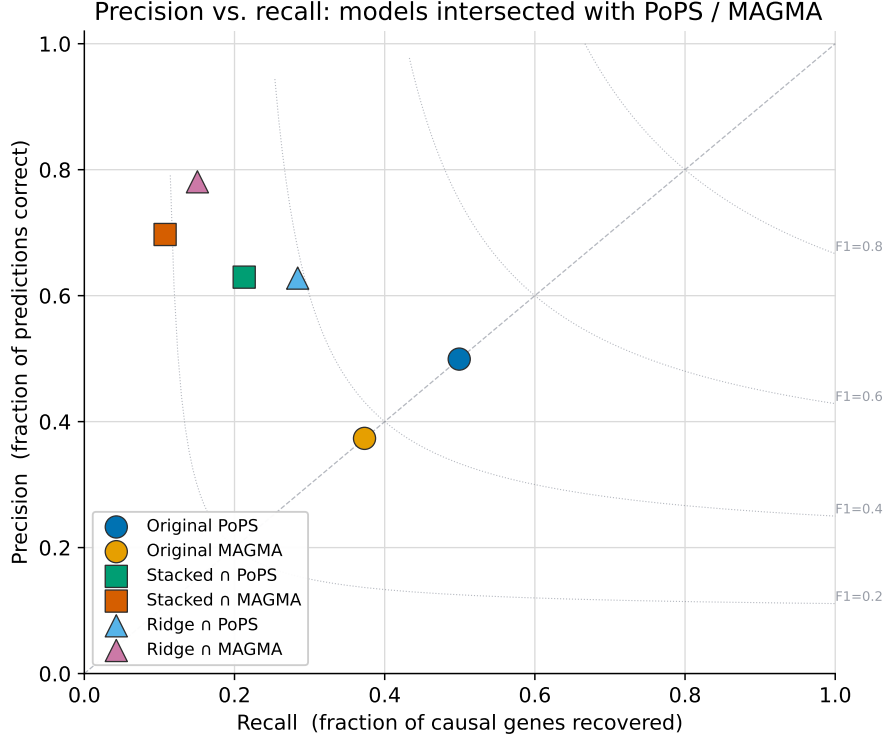


Figure 1: Precision vs. recall. Baselines (circles) lie on the diagonal; intersections (squares/triangles) lie above it, at higher precision and lower recall. Dotted curves are iso- F_1 contours.

8 Discussion

Three findings stand out. First, *model class dominated*: ridge on PoPS nearly doubled the random forest on identical features and recovered $\sim 87\%$ of PoPS, confirming that the estimator, not the sequence features, limited the initial model. Second, *AlphaGenome is redundant in the pooled metric* once a strong linear model has the PoPS features ($+0.002$; 6/6/33 per trait), so its apparent random-forest-era value was a weak-baseline artifact. Third, and countervailing, *the sequence signal is real and trait-specific*: on quantitative serum biomarkers with local cis-regulation our models exceed PoPS, and the stacking weights independently rediscover this class. Integration strategy also mattered—late fusion beat early fusion, and un-scaled concatenation was actively harmful—underscoring that *how* heterogeneous features are combined can matter as much as which features are used.

9 Future scope

- **True PoPS with the LD term.** Add PoPS’s gene–gene covariance (GLS whitening) to close the residual gap from 0.436 to 0.499.
- **Per-trait / per-class fusion weights.** A single global weight is provably suboptimal; adaptive or confidence-gated fusion should dominate both fusions.
- **Principled stacking.** Non-negative or fully cross-fitted meta-learners to respect the dependence between base scores.

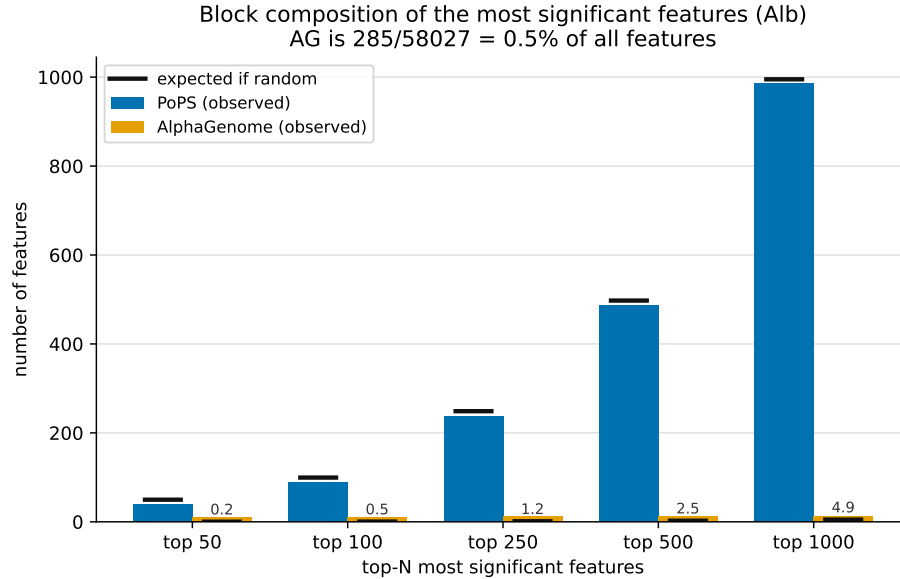


Figure 2: PoPS (blue) vs. AlphaGenome (orange) counts among the top- N most significant Albumin features; black dashes mark the count expected if features were selected at random. AlphaGenome sits far above its chance share throughout the top of the ranking.

- **Richer AlphaGenome signal.** Extend beyond Δ RNA to splicing, chromatin state and TF binding, and recover the $\sim 79\%$ of variants currently dropped for lack of CADD.
- **Stricter evaluation.** Leave-one-chromosome-out scoring and less-regularised forests as conservative robustness checks.

References

- [1] C.A. de Leeuw *et al.* MAGMA: generalized gene-set analysis of GWAS data. *PLoS Comput. Biol.* 11(4):e1004219, 2015.
- [2] E. M. Weeks *et al.* Leveraging polygenic enrichments of gene features to predict genes underlying complex traits and diseases. *Nature Genetics* 55:1267–1276, 2023.
- [3] Ž. Avsec *et al.* AlphaGenome: advancing regulatory variant-effect prediction with a unified DNA sequence model. Google DeepMind, 2025.
- [4] P. Rentzsch *et al.* CADD: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Res.* 47(D1):D886–D894, 2019.
- [5] L. Breiman. Random forests. *Machine Learning* 45(1):5–32, 2001.
- [6] A. E. Hoerl and R. W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics* 12(1):55–67, 1970.
- [7] D. H. Wolpert. Stacked generalization. *Neural Networks* 5(2):241–259, 1992.
- [8] Finucane Lab. Data release. <https://www.finucanelab.org/data>.